

Ozan Malcı

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Education

Sakarya University, BSc in Computer Engineering – Sakarya, Türkiye

Sept 2021 – July 2026

- Selected Coursework
 - Introduction to Cryptology – classical and modern crypto primitives, symmetric and public-key systems
 - Network Programming – socket APIs, concurrency, client-server systems
 - System Administration – server management, virtualization, network services
- Capstone Project: **SDN-Based Automated Threat Detection and Mitigation System**

Projects

P2P File Exchange – github.com/ozan2003/p2p_file_exchange

Designed and implemented a secure peer-to-peer file transfer desktop application for local networks using C# and Avalonia UI.

- Implemented UDP-based peer discovery with Ed25519-signed announcements to mitigate impersonation attacks.
- Built end-to-end encrypted file transfers over TCP using X25519 key exchange and ChaCha20-Poly1305 authenticated encryption.
- Implemented trust-on-first-use identity verification backed by a persistent **SQLite** trust database.

RFC Reader – github.com/ozan2003/rfc_reader

Developed a **Rust** terminal application for browsing IETF RFCs using a TUI built with Ratatui, optimized for keyboard-driven navigation.

- Implemented an HTTP fetch and persistent on-disk caching layer to support offline mode, reduce repeated network calls, and improve document load latency.
- Built full-text search over the document with in-app highlighting/navigation to accelerate lookup of protocol keywords and sections.

Heartbeat Monitor – github.com/ozan2003/heartbeat_monitor

Built a lightweight **Python** client-server monitoring system that runs configurable probes to check endpoint reachability/health across a network and aggregates telemetry centrally.

- Implemented a scheduled polling/heartbeat model with timeouts and failure detection to capture latency and outage events, including missed heartbeats indicative of packet-loss-like conditions.
- Ingested probe telemetry over the network and stored normalized results in **SQLite** for history/trending.

Skills

Programming Languages: Python, Rust, C#, C++

Tools: GNU Make, pytest, Git, uv

Network Simulation: GNS3, Cisco Packet Tracer

RDBMS: SQLite

Languages: Turkish (native), English